

Arkaprava Chakraborty

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SKILLS

- **Languages:** Python, Java
- **Frameworks:** TensorFlow, MediaPipe, PyTorch
- **Tools/Platforms:** OpenCV, Scikit-Learn, Numpy, Pandas, PyAutoGUI, Tkinter, Matplotlib, JSON
- **Soft Skills:** Problem-Solving Skills, Leadership, Project Management, Adaptability

INTERNSHIP

- **Freelance** March 2025
 - *Self-Employed*
 - **About:** The AI-powered outfit recommendation system is a backend software using weather API for real-time personalized clothing suggestions. It applies a hybrid recommendation engine with content-based filtering, collaborative filtering, and multi-factor analysis, leveraging user preferences, fashion trends, and item attributes.
 - **Tech stacks used:** Tensorflow, Scikit-learn, PyTorch, FastAPI(Backend Framework)

PROJECTS

- **Neuro-Agentix: Hand-Gesture Based Windows Management System** Mar 2025 - Apr 2025

A Real-time, vision-based interface that allows users to control the computer cursor using only finger movements, eliminating the need for a physical mouse. The system detects hand gestures, maps finger positions to screen coordinates, and interprets gestures for click, scroll, and drag operations. It combines computer vision, AI, and GUI automation to deliver smooth, responsive, and accurate cursor control. The system was trained and optimized to reduce lag and increase precision, enabling natural and intuitive human-computer interaction.
Tech: Python, Tensorflow, MediaPipe, OpenCV, Numpy, PyAutoGUI
- **AEDPA: Asteroid Early Detection and Preparedness Assistant** Jun 2025 - Aug 2025

A Python-based desktop application for monitoring near-Earth objects. Features real-time NEO tracking, threat assessment, and emergency protocols. Built with CustomTkinter for a modern UI, it integrates with NASA's NEO API. Includes dashboard visualization, hazard classification, and alert systems. Designed for both public awareness and educational use, providing comprehensive asteroid data and impact analysis in an accessible interface.
Tech: Python, Pillow, Requests(NASA's NEO API), JSON, CustomTkinter, datetime, os
- **Trackelectro: A desktop-based CV application to track electronic devices** Feb 2025

A Python-based computer vision application using OpenCV and NumPy for real-time electronics device tracking. Features multi-camera support, device detection, and timestamp logging. Core functionalities: camera feed processing, device recognition, and data logging. Built with Python, compatible with Windows, and utilizes threading for concurrent operations. Ideal for inventory management and device monitoring.
Tech: Python, OpenCV, Numpy, datetime, Threading
- **VigilEye Proctor: An Exam Proctored Camera** Aug 2025 - Sep 2025

A Python-based proctored-exam camera that leverages OpenCV for real-time video capture, Haar-cascade face detection, and YOLOv8n from Ultralytics for lightweight object detection. The system registers the examinee's face, tracks identity, and flags anomalies—missing face, multiple faces, mobile phone presence—saving violation images and issuing audible warnings via pyttsx3. Numpy supports vectorized face-encoding comparisons, while time-stamped cooldown logic throttles alerts. Directory management stores evidence under "violations".
Tech: Python, OpenCV, Numpy, ultralytics (YOLOv8 i.e YOLO nano), pyttsx3
- **A.A.T.M.A algorithm-based CPU-Cooling Agent Simulator (under research)** Apr 2025-cont.

The CPU Cooling Agent Simulator is a Python-based desktop utility built with Tkinter, running on my proprietary A.A.T.M.A algorithm to deliver predictive cooling control. Designed as a lightweight simulator, it integrates Matplotlib and NumPy for responsive, real-time graphs using FigureCanvasTkAgg, while psutil provides CPU usage, battery status, and frequency monitoring. Efficient time-series data handling is achieved with deque, and hardware

communication is enabled through WMI, pywin32, and ctypes, interfacing with OpenHardwareMonitor for fan-speed control. The A.A.T.M.A algorithm, powered by scikit-learn's LinearRegression, forecasts thermal trends for proactive cooling adjustments. Background threading ensures smooth updates synchronized with time and datetime, delivering a Windows-native simulator that intelligently demonstrates real-time monitoring, thermal prediction, cooling profiles, and user-friendly visualization.

A.A.T.M.A algorithm (hypothetical till yet), an Arkaprava's Adaptive Thermal-Management Algorithm, ingests live CPU metrics via psutil, buffers them in a deque, and forecasts ten-second temperatures through scikit-learn linear regression. Dynamic thresholds shift with user profiles, driving WMI-controlled fan curves. Health scoring, power estimation, quick-cool overrides, and battery-aware downgrades complete a proactive, self-tuning cooling loop.

Tech: Python, Numpy, psutil, matplotlib, wmi (via OpenHardwareMonitor via license), pywin32, ctypes, scikit-learn

• **Gold and Stock Market Probabilistic Prediction Model**

Sept 2025 - Dec 2025

This ML pipeline is a complete probabilistic forecasting pipeline built using Python 3.12, incorporating PyMC for Bayesian analysis, scikit-learn and XGBoost for conventional machine learning, TensorFlow/Keras for deep learning (LSTM), and statsmodels(ARIMA)/Prophet for time series analysis. This software retrieves financial data with yfinance, derives technical indicators (RSI, MACD, volatility) with ta, and orchestrates the entire process with a Tkinter GUI. The most fascinating aspect is its Bayesian model, which quantifies uncertainty with MCMC sampling, and its hybrid strategy that combines probabilistic forecasting with ML predictions to generate risk-aware trading signals. All results are evaluated with walk-forward backtesting, yielding Sharpe ratios, drawdowns, and directional accuracy, with results saved as .npy, .json, and .png files. Written by me, this software is a production-ready, end-to-end solution for gold and stock market prediction that is statistically valid and grounded in the latest research in ML.

Tech: PyMC, scikit-learn, XGBoost, TensorFlow/Keras, statsmodels, Prophet, yfinance, ta, and Tkinter

CERTIFICATES

- Computer Communications Specialization

Sept 2024 - Nov 2024
- Prompt Engineering Specialization

Dec 2024
- Scientific Computing in Python

Jan 2024

ACHIEVEMENTS

- Had the opportunity to educate and mentor orphan children at *Andal Parivar Orphanage* on computer fundamentals, digital safety, and basic artificial intelligence concepts. Conducted interactive sessions to help them understand how to use computers responsibly, identify online frauds, and gain awareness about AI and its real-world applications, fostering curiosity and digital empowerment among young learners.

EDUCATION

- Lovely Professional University**

• *Bachelor of Technology - Computer Science and Engineering*

Punjab, India

Since Aug 2023
- St. Patrick's H.S School**

• *High School (10+2)*

Asansol, West Bengal

Apr 2021 - Mar 2023